

Case Title

SSH Brute-Force Attack Investigation

Case Reference

DFIR-NET-SSH-001

Analyst

Denzel T Mchawaya
Cyber Security & Digital Forensics Analyst (Student)

Date of Analysis

23/12/2025

1. Executive Summary

This report documents the forensic analysis of captured network traffic related to suspicious SSH activity targeting an Ubuntu Linux server. The investigation was conducted to determine the nature of the activity, identify the source of the attack, and assess potential security implications.

Analysis of the packet capture (PCAP) file revealed evidence consistent with an **automated SSH brute-force or enumeration attack** originating from a single internal IP address. The attack involved repeated TCP connection attempts to the SSH service (port 22), utilization of an automated SSH client library, and rapid session termination patterns typical of credential-guessing tools.

No evidence of successful authentication was observed during the analyzed timeframe.

2. Scope of Investigation

The scope of this investigation was limited to:

Analysis of a single packet capture (PCAP) file

Identification of network-level indicators of compromise

Determination of attack type and attacker behavior

No host-based malware analysis was performed at this stage

3. Evidence Collected

Evidence Type	Description
PCAP File	Network traffic capture containing SSH-related communications
Capture Tool	tcpdump
Analysis Tool	Wireshark
Protocols Observed	TCP, SSHv2

4. Environment Overview

Role	System	IP Address
Attacker	Kali Linux	192.168.56.102
Target	Ubuntu Server (OpenSSH)	192.168.56.103
Network	Host-only virtual network	192.168.56.0/24

5. Methodology

The forensic analysis followed standard digital forensics procedures:

Preservation of captured network traffic

Inspection of TCP connection behavior

Protocol-level analysis of SSH traffic

Identification of anomalous patterns and automation indicators

Correlation of events over time

Wireshark display filters were applied to isolate SSH traffic and TCP handshake behavior.

6. Detailed Findings

6.1 Reconnaissance Activity

Initial packets revealed multiple **TCP SYN packets** sent from the attacker to the target's SSH port (TCP/22), followed by immediate TCP RST responses. This pattern is consistent with **half-open TCP SYN scanning**, commonly used to identify open ports without establishing full connections.

Finding:

The attacker confirmed the availability of the SSH service prior to exploitation attempts.

6.2 Establishment of SSH Sessions

Subsequent traffic demonstrated repeated **full TCP three-way handshakes**, indicating deliberate attempts to establish SSH connections. Each session was short-lived and followed a repetitive pattern.

Finding:

The attacker transitioned from reconnaissance to exploitation attempts.

6.3 Automated Client Identification

SSH protocol banners revealed the client software string:

SSH-2.0-libssh_0.11.3

This client library is commonly used in automated attack tools and scripts rather than interactive human-driven SSH sessions.

Finding:

The attack was conducted using an automated tool rather than manual login attempts.

6.4 Repeated SSH Key Exchange Attempts

Multiple SSH sessions displayed repeated key exchange initiation (Elliptic Curve Diffie-Hellman), followed by encrypted packet exchange and abrupt termination. These sessions were repeated rapidly across different source ports.

Finding:
The repeated incomplete SSH handshakes indicate failed authentication attempts typical of brute-force or credential enumeration attacks.

6.5 Port and Session Behavior

The attacker utilized a high number of ephemeral source ports when connecting to the SSH service, suggesting an attempt to increase attack speed and evade simple connection-based detection mechanisms.

Finding:
Connection behavior aligns with known brute-force attack patterns.

7. Indicators of Compromise (IOCs)

Indicator Type	Value
Attacker IP	192.168.56.102
Target IP	192.168.56.103
Target Service	SSH (TCP/22)
Client Identifier	SSH-2.0-libssh_0.11.3
Attack Type	SSH Brute Force / Enumeration
Evidence Type	Network PCAP

8. Assessment & Impact

No successful authentication was observed in the captured traffic

The SSH service was actively targeted

Continued attacks of this nature could lead to credential compromise if weak passwords exist

The activity poses a **medium to high risk** if left unmitigated

9. Recommendations

Enforce strong password policies and disable password-based SSH authentication

Implement SSH key-based authentication

Restrict SSH access using firewall rules or IP allowlists

Deploy intrusion detection systems (e.g., Fail2Ban, Zeek)

Monitor authentication logs for repeated failures

Integrate SSH logs into a SIEM platform for real-time alerting

10. Conclusion

The analyzed network traffic conclusively demonstrates an **automated SSH brute-force attack** originating from an internal network host. The attack was detected at the network level prior to successful authentication, highlighting the importance of continuous traffic monitoring and forensic readiness.

This investigation validates the effectiveness of packet capture and analysis as a core component of cyber security incident response.

Prepared By

Denzel T Mchawaya
Cyber Security & Digital Forensics Analyst
Zimbabwe